

Curriculum-Vitae of Pramod K. Mishra

Postal Address:

Dr. Pramod Kumar Mishra,
Senior Member, IEEE(USA), Fellow Institute of Mathematical
Analysis(UK)

Professor & Ex-Head

Dept. of Computer Science
(Institute of Eminence)

Banaras Hindu University

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Academic Training:

Post Doc:

In parallel and distributed computing: Indian Institute of Technology, Kharagpur,
Parallel and distributed complexity of shortest path algorithms” Dept. of
Electronics & Electrical Communication Engineering, Indian Institute of Technology,
Kharagpur, WB (India), 2005.

**Ph.D.: “A study of an efficient shortest path algorithms for serial and parallel
computers”**

All Examinations passed in one attempt.

Areas of Interest

AI and Machine Learning Algorithms, Data Analytics, Parallel Computing, High Performance
Clusters, Algorithm Engineering (AE), High Performance AE, Parallel Computation, and
Computational complexity,

Summary of Research

I study fundamental issues in the design and analysis of algorithms for combinatorial problems and in computational complexity. The main focus of my current research is parallel algorithm design which includes the design of efficient algorithms and highly parallel algorithms for graph problems, the evaluation for parallel computation, and the implementation of algorithms on parallel machines. A major focus is on the design and analysis of efficient algorithms for solving fundamental computational problems, where efficiency can be measured in terms of different resources such as time, space, number of processors, and number of random bits.

Research Interests

My research interests are in algorithms and computational theory primarily in the areas of graph algorithms, data structures, randomised algorithms, parallel algorithms and parallel data mining algorithm. My primary research interest is on algorithm design, and my current research explores three avenues: Design and analysis of efficient sequential, parallel and distributed algorithms for different types of problems; effective modelling of parallel algorithm design; and experimental evaluation of algorithms, AI based algorithms.

7. Awards/ Fellowships Received:

1. Selection in INSA Bilateral Exchange Programme to visit Germany for 4 weeks, from April 2011- March, 2012.
2. Max Planck Institute Fellowship, MPI Saarbrucken, Germany, 2011 Research Collaborator: Prof Kurt Mehlhorn, Director, MPI.
3. UGC nomination to visit **Mauritius under 4th UGC-TEC Consortium 2007-2010** for three months under exchange programme.
4. Elected as a **Fellow of Institute of Mathematical Sciences and Applications, FIMA(UK)** 2009 for outstanding contribution in parallel and distributed computation.
5. INSA award of Visiting Scientist Fellowship, Indian National Academy of Science, New Delhi, 2005-2006.
6. Max Planck Institute, Bonn (Germany), Travel Grant Fellowship, 2006.
7. Armond Borel Foundation Fellowship, Switzerland, 2002.
8. HiPC Travel Grant Fellowship, Goa, 2005.

9. TIFR Visiting Research Fellowship, Tata Institute of Fundamental Research Colaba, Bombay, 2005.
10. Visiting Research Scientist Fellowship, INRIA France, 1999-2000.
11. Nomination in the 22nd edition of America publisher Marquis Whos Who in the world, 2005.
12. Nomination in the 23rd edition of America publisher Marquis Whos Who in the world, 2006.
11. Nomination in the Dictionary of International Biography, Cambridge, England 2005.

Academic Visit:

- Swiss Federal Institute of Technology (*Eidgenössische Technische Hochschule*) Zürich, 2002.
Host: Prof Jens Gustedt Currently Professor at INRIA, France.
- Max Planck Institute for Mathematics (MPI), Bonn, Germany, 2005.
Host: Prof Matilde Marcolli, Currently Professor at California Institute of Technology, Caltech, USA.
- University of Leipzig, Germany, 2010.
Host: Prof Manfred Droste
- Max Planck Institute for Informatics (MPI), Saarbrücken, Germany, 2011. Host: Prof Kurt Mehlhorn, Director, MPI
- University of Valencia, Spain.
- University of Science & Technology, Hefei (China).

8. Project:

- DAE Project: Completed: Amount: INR 5, 83, 200.00
- UGC Project: Completed: Amount: INR 80,000.00
- CSIR Project: Running: Amount: INR 8,83,261.00
- DST-CIMS: 6.25 Cr (Sanctioned Nov 20, 2007, Started July 2008 (2008-2013): Ref: SR/S4/MS-428/07
- DST-PURSE: 20 Lacks (2009-2012)
(Establishment of High Performance Computing Lab)

- TWAS: Project submitted: USD 6000.00

9. Visiting Faculty:

Indian Railway Institute of Mechanical and Electrical Engineering (**IRIMEE**), Jamalpur, Bihar (India).

10. Professional & Academic Qualification:

(a) Professional Experience:

1. Paper setter and Examiner of the University Education, in the field of Mathematical and Computational Sciences.
2. *Group Leader & PI of Theoretical Computer Science, DST Centre for Interdisciplinary Mathematical Sciences, BHU*
3. Reviewer of the various National & International Journals of repute.
 - (i) Applied Mathematics Letters, Elsevier Science, The Netherlands
 - (ii) Information Science, The Netherlands
 - (iii) Chaos and Complexity Letters, USA
 - (iv) Journal of Zhn Univ Science (Springer)
 - (v) Discrete Mathematics Elsevier Science.
 - (vi) IEEE Transaction on Knowledge and Engineering, USA
 - (vii) IEEE Transaction on Parallel and Distributed Systems, USA
 - (viii) Theoretical Computer Science, Elsevier, The Netherlands
 - (ix) Journal of Computational Science, Elsevier.
4. Assistant Editor, IJED (An International Journal of IIT-Roorkee)
{From January 2003-August, 2004.}
5. Associate Editor, IJED (An International Journal of IIT -Roorkee) **{From September, 2004-January, 2006}.**
6. Moderator, IJED (An International Journal of IIT -Roorkee)
{From January, 2006- to date}.

(b) Administrative Experience:

- a. Head, Department of Computer Science, Banaras Hindu University, Varanasi (01.04.2013- 31.03.2016.) (Second term)
- b. Head, Dept of computer Science, Banaras Hindu University, Varanasi wef 01.04.2007 to 31.03.2010.
- c. Co-ordinator, MCA Programme, RGSC, Barkachha, Mirzapur wef 01.04.2007 to Continuing.

- d. Head-in-Charge, Dept. of Computer Science, BHU – Varanasi, on: 14.12.2006, 20-12-2006 to 24-12-2006, 01.10.2006-09.10.2006, 26.02.2007-01.03-2007
- e. Professor in Charge, Computer Centre, Dept. of Applied Mathematics, Birla Institute of Technology, Mesra (India) from: 01.04.2000-06.12.2005.
- f. Param 10000 Super Computing Lab, Birla Institute of Technology, Mesra (India) from: 01.04.2000-06.12.2005.
- g. Professor-in Charge, HPC Research Group, Birla Institute of Technology, Mesra (India) from: 01.04.2000-06.12.2005.

12. Conference /Workshop/Symposia Organized:

- National Conference and Workshop on High Performance Computing and Applications, Department of Computer Science, Banaras Hindu University, Varanasi, February, 25-27, 2008.
- All India Computer Science Student Technical Meet, TORQUE-2008, Department of Computer Science, Banaras Hindu University, Varanasi, February 25-27, 2008.
- Workshop on Discrete Mathematics & Applications, Department of Computer Science, Banaras Hindu University, Varanasi, February 25-26, 2008.
- National Conference and Workshop on High Performance Computing and Applications, Department of Computer Science, Banaras Hindu University, Varanasi, February, 09-11, 2009.
- All India Computer Science Student Technical Meet, TORQUE-2009, Department of Computer Science, Banaras Hindu University, Varanasi, February 09-11, 2009
- Research Promotion Workshop on Introduction to Graph and Geometric Algorithms, ***Jointly with Prof Subir Ghosh, School of Technology and Computer Science, Tata Institute of Fundamental Research Bombay, January 27-29, 2010.***
- National Conference and Workshop on High Performance Computing and Applications, Department of Computer Science, Banaras Hindu University, Varanasi, February, 08-10, 2010.

- All India Computer Science Student Technical Meet, TORQUE-2010, Department of Computer Science, Banaras Hindu University, Varanasi, February 10-11, 2010.
- Research Promotion Workshop on Introduction to Graph and Geometric Algorithms(IGGA-2010), Department of Computer Science, Banaras Hindu University, Varanasi, February 08-13, 2010.
- National Instructional Workshop on Graph Theory and Applications (WGTA-2011), May 23-28, 2011, DST Centre for Interdisciplinary Mathematical Sciences & Department of Computer Science, Banaras Hindu University, Varanasi.
- Workshop on “Contribution of Mahamana Pd Mdan Mo han Malviya Ji “, 2010-2011
- Symposia on “ Mahamana a great visionary” 2010-2011
- Seminar “A play on Malviya ji” 2010-2011
- Research Promotion Workshop on Introduction to Graph Algorithms and Applications (IGGA-2012), Department of Computer Science & DST-CIMS, Banaras Hindu University, Varanasi, November 05-10, 2012.
- National Conference and Workshop on High Performance Computing and Applications, Department of Computer Science, Banaras Hindu University, Varanasi, February, 05-07, 2012.
- Refresher Course on ICT Applications, Academic Staff College, Banaras Hindu University, Varanasi, September 22- Oct 12, 2012.

13. M. Tech Students Guided: 21

14. Ph. D. Students Guided:15

15. Ph D students registered: 8

16. List of significant publications:

Journals

2022

1. Sharma, A., & Mishra, P. K. (2022). Covid-MANet: Multi-task attention network for explainable diagnosis and severity assessment of COVID-19 from CXR images. *Pattern Recognition*, 131, 108826. [SCI Indexed]
2. Sharma, A., Mishra, P.K. Image enhancement techniques on deep learning approaches for automated diagnosis of COVID-19 features using CXR

images. *Multimedia Tools and Applications* **81**, 42649–42690 (2022).
<https://doi.org/10.1007/s11042-022-13486-8>

3. Ajay Sharma, P K Mishra, Deep Learning Approaches for Automated Diagnosis of COVID-19 Using Imbalanced Training CXR Data: Advanced Network Technologies and Intelligent Computing Vol 131, pp 453–472 (2022)
<https://doi.org/10.1016/j.patcog.2022.108826>.

4. Sharma, A., & Mishra, P. K. (2021). Performance analysis of machine learning based optimized feature selection approaches for breast cancer diagnosis. *International Journal of Information Technology*, 1-12.

5. Srivastava, A., & Mishra, P. K. (2022). Multi-attributes based energy efficient clustering for enhancing network lifetime in WSN's. *Peer-to-Peer Networking and Applications*, 1-24.

6. Srivastava, A., & Mishra, P. K. (2022, February). Multiple-Parameter Based Clustering for Efficient Energy in Wireless Sensor Networks. In *Advanced Network Technologies and Intelligent Computing: First International Conference, ANTIC 2021, Varanasi, India, December 17–18, 2021, Proceedings* (pp. 15-24). Cham: Springer International Publishing.

7. Sadhukhan, S., & Mishra, P. K. (2022). A multi-layered hybrid model for cancer cell invasion. *Medical & Biological Engineering & Computing*, 60(4), 1075-1098.

8. Sadhukhan, S., & Mishra, P. K. (2022). The notion of fractals in tumour angiogenic sprout initiation model based on cellular automata. *Chaos, Solitons & Fractals*, 155, 111717.

9. Sadhukhan, S., Mishra, P. K., Basu, S. K., & Mandal, J. K. (2021). A multi-scale agent-based model for avascular tumour growth. *Biosystems*, 206, 104450.

10. S Ketu, PK Mishra, A Contemporary Survey on IoT Based Smart Cities: Architecture, Applications, and Open Issues, *Wireless Personal Communications*, 1-49 (2022).

11. S Ketu, PK Mishra, Hybrid classification model for eye state detection using electroencephalogram signals *Cognitive Neuro Dynamics* 16 (1), 73-90 (2022).
12. S Ketu, PK Mishra, Empirical analysis of machine learning algorithms on imbalance electrocardiogram based arrhythmia dataset for heart disease detection, *Arabian Journal for Science and Engineering* 47 (2), 1447-1469 (2022).
13. S Ketu, PK Mishra, India perspective: CNN-LSTM hybrid deep learning model-based COVID-19 prediction and current status of medical resource availability, *Soft Computing* 26 (2), 645-664 (2022).
14. Kushwaha, B. S., & Mishra, P. K. (2022). A Survey on Cross-Layer Optimization in Wireless Networks. *J. Adv. Comput. Netw*, 10, 1-9.
15. Srivastava, M., Srivastava, A. K., Garg, R., & Mishra, P. K. (2022). Performance Evaluation of the MapReduce-based Parallel Data Pre-processing Algorithm in Web Usage Mining with Robot Detection Approaches. *IETE Technical Review*, 39(4), 865-879.

2021

16. Shwet Ketu and **P. K. Mishra**. "Cloud, Fog and Mist Computing in IoT: An Indication of Emerging Opportunities." *IETE Technical Review*, 1-12, (2021). (SCI Indexed) IF:1.845
17. Srivastava, A., & Mishra, P. K. (2021). A survey on WSN issues with its heuristics and meta-heuristics solutions. *Wireless Personal Communications*, 121(1), 745-814.
18. Mishra, A., Mishra, K. S., & Mishra, P. K. (2021). Performance evaluation of simulated annealing-based task scheduling algorithms. In *Information Management and Machine Intelligence: Proceedings of ICIMMI 2019* (pp. 145-152). Springer Singapore.

19. Pandey, N., & Mishra, P. K. (2022, February). A Survey on DDoS Attacks on Network and Application Layer in IoT. In *Advanced Network Technologies and Intelligent Computing: First International Conference, ANTIC 2021, Varanasi, India, December 17–18, 2021, Proceedings* (pp. 240-250). Cham: Springer International Publishing.

2020

20. Shwet Ketu and **P. K. Mishra**. "A Hybrid Deep Learning Model for COVID-19 Prediction and Current Status of Clinical Trials Worldwide." *Computers, Materials & Continua* 66 (2), 1896-1919, (2020). (SCI Indexed) IF:4.89

21. Shwet Ketu and **P. K. Mishra**. "Enhanced Gaussian process regression-based forecasting model for COVID-19 outbreak and significance of IoT for its detection." *Applied Intelligence* 51, 1492–1512, (2020). (SCI Indexed) IF:3.325

22. Shwet Ketu, **P. K. Mishra** and Sonali Agarwal. "Performance Analysis of Distributed Computing Frameworks for Big Data Analytics: Hadoop Vs Spark." *Computación y Sistemas* 24 (2), (2020). (Scopus Indexed)

23. Shwet Ketu and **P. K. Mishra**. "Performance Analysis of Machine Learning Algorithms for IoT-Based Human Activity Recognition." *Advances in Electrical and Computer Technologies*, 579-591, (2020). (Scopus Indexed)

24. Radharaman Chandan and **P. K. Mishra**. "FSM based Intrusion Detection of Packet Dropping Attack using Trustworthy Watchdog Nodes." *Recent Patents on Computer Science* 13, (2020).

25. Radharaman Chandan and **P. K. Mishra**. "Consensus Routing And Environmental Discrete Trust Based Secure AODV in MANETs." *International Journal of Computer Networks & Communications (IJCNC)* Vol 12, (2020).

26. **Sharma, A.**, & Mishra, P. K. (2020). State-of-the-art in performance metrics and future directions for data science algorithms. *Journal of Scientific Research*, 64(2), 221-238. (Scopus Indexed)

27. Pandey, N., & Mishra, P. K. (2020). Taxonomy of DDoS attacks and their defense mechanisms in IoT. *Journal of Scientific Research*, 64(2).

2019

28. Ankita Srivastava and **P. K. Mishra**. "State-of-the-art Prototypes and Future Propensity Stem on Internet of Things." *International Journal of Recent Technology and Engineering (IJRTE)* 8 (4), (2019). (Scopus Indexed)

29. Pankaj Singh, Sudhakar Singh, **P. K. Mishra**, and Rakhi Garg. "A data structure perspective to the RDD-based Apriori algorithm on Spark." *International Journal of Information Technology* (2019): 1-10.

30. Atul Kumar Srivastava, Mitali Srivastava, Rakhi Garg, and **P. K. Mishra**, "An Aitken-extrapolated Gauss-Seidel method for computing PageRank." *Journal of Statistics and Management Systems* 22.2 (2019): 199-222.

2018

31. Radharaman Chandan and **P. K. Mishra**. "A Review of Security Challenges in Ad-Hoc Network." *International Journal of Applied Engineering Research* 13 (22), 16117-16126, (2018).
32. Bindeshwar Singh Kushwaha and **P. K. Mishra**. "Cross-Layer Effect of Transmit Power on the MAC, the Network and the Transport Layer in Mobile Ad Hoc Network." *International Journal of Computer Networks and Wireless Communications (IJCNCW)* 8 (5), 1-8, (2018).
33. Radharaman Chandan, Bindeshwar Singh Kushwaha and **P. K. Mishra**. "Performance Evaluation of AODV, DSDV, OLSR Routing Protocols using NS-3 Simulator." *International Journal of Computer Network & Information Security* 10 (7), (2018).
34. Atul Kumar Srivastava, Rakhi Garg, and **P. K. Mishra**, "Analysis of Iterative Methods in PageRank Computation" *Journal of Information and Optimization Sciences*, Taylor & Francis, ISSN: Print ISSN: 0252-2667 Online ISSN: 2169-0103, Volume 39, Issue 1, Page No: 129-142, <https://doi.org/10.1080/02522667.2017.1372914>, 2018. . (Web of Science (ESCI))
35. Mitali Srivastava, Rakhi Garg, and **P. K. Mishra**, "A MapReduce-based User Identification Algorithm in Web Usage Mining" *International Journal of Information Technology and Web Engineering*, IGI Global, ISSN: Print ISSN: 1554-1045 Online ISSN: 1554-1053, Volume 13 Issue 2, Page No: 11-23, 2018. (Web of Science (ESCI), SCOPUS indexed)

2017

36. Mitali Srivastava, Atul Kumar Srivastava, Rakhi Garg, and **P. K. Mishra**, "Experimental Study of Time oriented and Referrer oriented Session Identification Methods in Web Usage Mining" *International Journal of Electronics Engineering*, ISSN: 0973-7383, 2017. (ICI Indexed)
37. Mitali Srivastava, Atul Kumar Srivastava, Rakhi Garg, **P. K. Mishra**, "Analysis of Robot Detection Approaches for Ethical and Unethical Robots on Web Server Log" *International Journal of Advanced Research in Computer Science*, ISSN: 0976-5697, Volume 8, No. 5, Page No: 1132-1134, 2017.
38. Mitali Srivastava, Rakhi Garg, and **P. K. Mishra**, "A MapReduce-based Parallel Data Cleaning Algorithm in Web Usage Mining" *International Journal of Computer Science and Applications (IJCSA)*, ISSN: 0972-9038, Volume 14, Issue 2, Page No: 115-129, 2017. (SCOPUS Indexed).
39. Atul Kumar Srivastava, Rakhi Garg, and **P. K. Mishra**, "Discussion on Damping Factor value in PageRank Computation" *International Journal of Intelligent Systems and Applications*, ISSN: 2074-904X (Print), ISSN: 2074-9058 (Online), Volume 9, Issue 9, Page No: 19-28, DOI: 10.5815/ijisa.2017.09.03, 2017. (SCOPUS Indexed).
40. Atul Kumar Srivastava, Mitali Srivastava, Rakhi Garg, and **P. K. Mishra**, "Parallel PageRank Algorithms: A Survey" *International Journal on Recent and Innovation Trends in Computing and Communication*, ISSN: 2321-8169, Volume: 5 Issue: 5, Page No: 470-473, 2017.

41. Sudhakar Singh, Rakhi Garg, **P. K. Mishra**, "Performance optimization of MapReduce-based Apriori algorithm on Hadoop cluster", Computers and Electrical Engineering (CAEE), Elsevier, ISSN: 0045-7906, In-Press, 2017, SCIE IF: 1.570.

2016

42. Bindeshwar Singh Kushwaha and **P. K. Mishra**. "Comprehensive analysis of ad hoc network routing protocols." International Journal of Computer Applications 139 (11), 1-5, (2016).
43. Bindeshwar Singh Kushwaha and **P. K. Mishra**. "Different Traffic Patterns Over Ad Hoc Network Routing Protocols." International Journal of Computer Applications 138 (11), 1-5, (2016).
44. Sudhakar Singh, Pankaj Singh, Rakhi Garg, **P K Mishra**, "Mining Association Rules in Various Computing Environments: A Survey", International Journal of Applied Engineering Research (IJAER), ISSN (Print): 0973-4562, ISSN (Online): 0973-9769, Vol. 11, No. 8, pp. 5629-5640, April 2016, Scopus Indexed.
45. Mishra, A., & Mishra, P. K. (2016). A randomized scheduling algorithm for multiprocessor environments using local search. *Parallel Processing Letters*, 26(01), 1650002.

2015

46. Sudhakar Singh, Rakhi Garg, **P K Mishra**, "Performance Analysis of Apriori Algorithm with Different Data Structures on Hadoop Cluster", International Journal of Computer Applications (IJCA), ISSN: 0975-8887, Vol. 128, No. 9, pp. 45-51, October 2015, DBLP Indexed.
47. Sudhakar Singh, Pankaj Singh, Rakhi Garg, **P K Mishra**, "Big Data: Technologies, Trends and Applications", International Journal of Computer Science and Information Technologies (IJCSIT), ISSN: 0975-9646, Vol. 6, No. 5, pp. 4633-4639, October, 2015.
48. Atul Kumar Srivastava, Mitali Srivastava, Rakhi Garg, **P. K. Mishra**, "Analysis of Successive over Relaxation method in PageRank Computation" International Journal of IT & Knowledge Management, ISSN: 0973-4414, Volume 9, Issue 1, Page No: 63-67, 2015.
49. Atul Kumar Srivastava, Mitali Srivastava, Rakhi Garg, **P. K. Mishra**, "Comparative Analysis of Power Method and Gauss-Seidel Method in PageRank Computation" International Journal of Computer Engineering and Applications, ISSN: 2321-3469, Volume 9, Issue 8, Page No: 1-10, 2015.
50. Atul Kumar Srivastava, Rakhi Garg, **P. K. Mishra**, "Issues in Sequential Web Page Ranking Algorithms" International Journal of Advanced Research in Computer Science and Software Engineering, ISSN: 2277-128X Volume 5 Issue 5, Page No: 1182-1186, 2015.
51. Mitali Srivastava, Atul Kumar Srivastava, Rakhi Garg, **P. K. Mishra**, "Analysis of User Identification Methods in Web Usage Mining" International Journal of Computer

Engineering and Applications, ISSN: 2321-3469 Volume 9, Issue 8, Page No: 57-65, 2015.

52. Mitali Srivastava, Atul Kumar Srivastava, Rakhi Garg, **P. K. Mishra**, "Comparative Analysis of Robot Detection Techniques on Web Server Log" International Journal of Advanced Research in Computer and Communication Engineering, ISSN: 2278-1021, Volume 4 Issue 9, Page No: 186-189, 2015.

2014

53. Mitali Srivastava, Rakhi Garg, and **P. K. Mishra**, "Preprocessing Techniques in Web Usage Mining: A Survey" International Journal of Computer Applications, ISSN: 0975-887, Volume: 97 Issue: 18, Page No: 1-9, 2014.

54. Atul Kumar Srivastava, Mitali Srivastava, Dr. Rakhi Garg, Prof. **P. K. Mishra**, "Comparative Study of Web Page Ranking Algorithms" International Journal of Emerging Technologies in Computational and Applied Sciences, ISSN (Online) 2279-0055 ISSN (Print) 2279-0047, Issue 7 volume 1 Page No: 26-32, 2014.

55. Sudhakar Singh, Rakhi Garg, **P. K. Mishra**, "A Comparative Study of Association Rule Mining Algorithms on Grid and Cloud Platform", International Journal of Emerging Technologies in Computational and Applied Sciences (IJETCAS), ISSN (Print): 2279-0047, ISSN (Online): 2279-0055, Vol. 2, Issue 7, pp. 127-134, March 2014, DBLP Indexed.

56. Hasan, M. M., & Mishra, P. K. (2013). Novel algorithm for skin color based segmentation using mixture of GMMs. *Signal & Image Processing*, 4(4), 139.

2012

57. **Mishra, P K**, Mishra, KS, Mishra, Abhisek, Tripathi, AK: A Randomized Scheduling Algorithm for Multiprocessor Environments, Parallel Processing Letters, Vol 22, Issue 4, 2012.

58. **Mishra P K**, Mishra, K S, Mishra Abhisek, Tripathi, A K: Benchmarking the clustering algorithms for multiprocessor environments using dynamic priority of modules, Applied Mathematical Modelling, Elsevier Vol 36, PP 6243-6263, 2012.

59. Ibrahim, N A, Hasan, M M, Khan, R Z, **Mishra, P K.**: Understanding Color Models: A review, ARPN, Journal of Sc & Tech, Vol 2, No 3, PP. 265-275, 2012.

60. Hasan,MM, Mishra, PK: Direction Analysis Algorithm using Statistical Approaches, Fourth International Conference on Digital Image Processing (ICDIP 2012), edited by Mohamed Othman, Sukumar Senthilkumar, Xie Yi, Proc. of SPIE Vol. 8334, 83340L ·©2012 SPIE, CCC code: 0277-786X/12/\$18 · doi: 10.1117/12.946046, Proc. of SPIE Vol. 8334 83340L-1-5.

61. Yadav, RS, **Mishra, PK**: Performance Analysis of High Performance k-Mean Data Mining Algorithm for Multicore Heterogeneous Compute Cluster, International Journal of Information and Communication Tech. Vol 2, No 4 PP. 390-394, 2012.

62. Hasan,MM, **Mishra, PK**: Hand gesture Modelling and Recognition using Geometric Features: A Review, Canadian Journal of Image Processing and Computer Vision, Vol.3,Issue(1),2012,pp 12-26.

63. Hasan,MM, **Mishra, PK**: Real Time Fingers and Palm Locating using Dynamic Circle Templates, International Journal of Computer Applications, Volume 41– No.6, PP 33-43, 2012.
64. Hasan,MM, **Mishra, PK**: Features Fitting using Multivariate Gaussian Distribution for Hand Gesture Recognition , Vol 3, No 2, PP 73-80, 2012.
65. Hasan,MM, **Mishra, PK** : Novel Algorithm for Multi Hand Detection and Geometric Features Extraction and Recognition, International Journal of Scientific & Engineering Research, Vol 3, Issue 5, PP 1-12, 2012.
66. Hasan,MM, **Mishra, PK** :Robust Gesture Recognition using Gaussian Distribution for Features Fitting, International Journal Machine Learning and Computing, Vol 2, No 3, PP 266-273, 2012.
67. Hasan,MM, **Mishra, PK**: Superior Skin Color Model using Multiple of Gaussian Mixture Model, British Journal of Science, Vol 6(1), PP 1-14, 2012.
68. Hasan,MM, **Mishra, PK**: Improving Morphology Operation for 2D Hole Filling Algorithm, International Journal of Image Processing, Vol.6,Issue(1),2012,pp 1-12.
69. Hasan,MM, **Mishra, PK**, : HSV brightness factor matching for gesture recognition system, International Journal of Image Processing, Vol.4,Issue(2),2012,pp 456-467.
70. Mishra, P. K., Mishra, A., Mishra, K. S., & Tripathi, A. K. (2012). Benchmarking the clustering algorithms for multiprocessor environments using dynamic priority of modules. *Applied Mathematical Modelling*, 36(12), 6243-6263.
71. Mishra, P. K., Mishra, K. S., Mishra, A., & Tripathi, A. K. (2012). A randomized scheduling algorithm for multiprocessor environments. *Parallel Processing Letters*, 22(04), 1250015.

2011

72. **Mishra P K**, Mishra, KS, Mishra, Abhishek: A Clustering Algorithm for Multiprocessor Environments using Dynamic Priority of Modules, Annales Mathematicae et Informaticae, Vol 38, pp. 99-110, 2011.
73. Garg, Rakhi, **Mishra P K**: Exploiting Parallelism in Association Rule Mining Algorithms, International Journal of Advancements in Technology, Vol 2, PP 222-232 (2011).
74. **Mishra PK** and Singh, Alok: Analysis of the randomized graph coloring algorithm for the optimal channel allocation in WLANS, International Journal of Contemporary Mathematical Sciences, USA, 2011.
75. Hasan,MM, **Mishra, PK**, : Brightness factor matching for gesture recognition system using scaled normalization, International Journal of Computer science & Info Tech Vol.3,Issue(2),2011,pp 35-46.
76. Hasan,MM, **Mishra, PK**,: Comparative study for construction of gesture recognition system, International Journal of Computer Sc & Software Tech. Processing, Vol.4,Issue(1),2011,pp 15-21.
77. Hasan,MM, **Mishra, PK**: Performance study for construction of gesture recognition system, International Journal of Computer Sc & Software Tech. Processing, Vol.4,Issue(1),2011,pp 25-28.
78. Hasan, M. M., & Mishra, P. K. (2011, June). Gesture recognition using modified HSV segmentation. In *2011 International Conference on Communication Systems and Network Technologies* (pp. 328-332). IEEE.

79. Mishra, P. K., Mishra, K. S., & Mishra, A. (2011, January). A clustering algorithm for multiprocessor environments using dynamic priority of modules. In *Annales Mathematicae et Informaticae* (pp. 99-110).

2010

80. Garg, Rakhi, **Mishra, P K**: Application of Data Mining in NREGA, CSI Communications, PP 154-165, (2010).
81. Yadav, RS and **Mishra, PK**: Use of MIS with ICT for making secure and transparent system for the management of NREGA scheme in rural India, CSI Communications, PP 125-136(2010).
82. **Mishra P K**, Mishra, KS, Mishra, Abhishek: A Clustering Heuristic for Multiprocessor Environments using Computation and Communication Loads of Modules, International Journal of Computer Science and Information Technology, Singapore Vol.2, No.5, PP 170-182, 2010.
83. Hasan, M. M., & Misra, P. K. (2010). Robust Gesture Recognition using Euclidian Distance. In *IEEE International Conference on Computer and Computational Intelligence, China* (Vol. 3, pp. 38-46).
84. Hasan, M. M., & Mishra, P. K. (2010). HSV brightness factor matching for gesture recognition system. *International Journal of Image Processing (IJIP)*, 4(5), 456-467.
85. Mishra, P. K., Mishra, K. S., & Mishra, A. (2010). A clustering heuristic for multiprocessor environments using computation and communication loads of modules. *International Journal of Computer Science & Information Technology*, 2(5), 170-182.

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2. **Yadav, RS, Mishra, PK**: Some observations of high performance data mining algorithms and challenges, Proc. of the International Conference on Advanced Computing and Communication Technologies (IACCT 2011) PP89-93, 2011.
3. **Mishra P K**: Complexity Analysis of Distributed Shortest Path Algorithms with MPI, International Conference on New Trends in Life Testing, Bayesian Inference, Sampling Theory, Bio Statistics, Bio Informatics and Computer Applications, Banaras Hindu University, P 67, January 5-8, 2011.
4. R S Yadav, **Mishra P K**: High Performance Statistical Data Mining Framework Architecture for Very Large Data Base, International Conference on New Trends in Life Testing, Bayesian Inference, Sampling Theory, Bio Statistics, Bio Informatics and Computer Applications, P 71, Banaras Hindu University January

5-8, 2011.

5. Garg Rakhi, **Mishra P K**: Data Mining: An Emerging Technology, International Conference on New Trends in Life Testing, Bayesian Inference, Sampling Theory, Bio Statistics, Bio Informatics and Computer Applications, P. 67, Banaras Hindu University January 5-8, 2011.
6. **Mishra, P K** :Complexity Analysis of Shortest Path Algorithms with High Performance Computing, Annual Session of the National Academy of Sciences, Jaipur National University, Jaipur, December 2-4, 2010.
7. Hasan, M M, **Mishra, P K**: Robust Gesture Recognition Using Euclidian Distance, ICCI, 2010, IEEE Explore, V3-38-V3-46.
8. Hasan, MM, **Mishra, PK**: A powerful and weakness points for gesture recognition techniques, 6th International Conference on Multimedia Information Technology and Applications MITA-2010, Kathmandu, Nepal, August 23-25, 2010, pp- 144-149.
9. Garg, Rakhi and **Mishra, PK**: High Performance data mining algorithm for the distributed relational database, **FDRCS-2012 -PP 48-50**.
10. Hasan, M M, **Mishra, P K**: Direction Analysis Algorithm using Statistical Approaches, Fourth International Conference on Digital Image Processing (ICDIP 2012), edited by Mohamed Othman, Sukumar Senthilkumar, Xie Yi, Proc. of SPIE Vol. 8334, 83340L · © 2012 SPIE,

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11. Hasan, M M, **Mishra, P K**: Rotation Variance Remedy for Gesture Recognition System, International Science and Technology Conference, Famagusta (China), Oct, 27-29,2010, PP 613-619.
12. Hasan, M M, **Mishra, P K**: Connected Boundary Region Tracing, for Object Hole Filling, International Conference on Advances in Engineering and Technology, 26th November, 2011, Bhubaneswar, PP 67-72.
13. Garg, Rakhi and **Mishra, PK**: *Some Observations of Sequential, Parallel and Distributed Association Rule Mining Algorithms*, **IEEE ICCAE-2009 PP 336-342**, 978-0-7695-3569-2/09 \$25.00 © 2009 IEEE, DOI 10.1109/ICCAE.2009.28.
14. Garg, Rakhi and **Mishra, PK**: *Issues and Challenges in Data Mining algorithms*, **IEEE IACC -2009, PP 942-946**.

Books Edited:

1. Basics of Object Oriented Programming & C++ Block -1, 4 Units, IGNOU, New Delhi, PP 01-116, 2012 (IGNOU- New Delhi- ISBN 978-81-266-6031-6)
2. Inheritance and Polymorphism in C++ Block -2, 3 Units PP 01-86, 2012 (IGNOU- New Delhi, ISBN 978-81-266-32-2).
3. Advanced Features of C++, Block -3, 4 Units , PP 01-74, 2012 (IGNOU- New Delhi- ISBN 978-81-6039-9)
4. Lab Manual- PP 01-21, 2012, (IGNOU New Delhi , ISBN 978-81-266-6036-0)

18. Workshops Attended/ Participated :

1. Workshop on **Parallel Computing Optimizing Performance of Parallel Programs**, C-DAC Pune, June 3rd -6th , 2002.
2. INDO-US Workshop on **High Performance Computing and Scientific Application-Future Trends**, Dec. 20, 2002, C-DAC, Bangalore, India.
3. Workshop on **Bio-Computing and Applications**, Dec 20-21, 2002, C-DAC Knowledge Park, C-DAC, Bangalore, India.
4. Short Term Training Program on **Parallel and Distributed Computing** (AICTE-ISTE Sponsored Program) CBIT Hyderabad, from June 24th -5th July, 2002.
5. Short Term Training Program on Conventional Energy (AICTE -ISTE Sponsored Program) Dept. of Mechanical Engineering, BIT- Mesra, from June 9th -25th June, 2003.

6. Borel symposium 2005, Zurich and Geneva, Switzerland June 26, 2005-July 06, 2005.

19. Invited Talk:

1. On Computer Algorithms for summarizing large data set: CMS, Pala, Kochi, April 5-8, 2010.
2. Computer Applications, International Conference on New Trends in Life Testing, Bayesian Inference, Sampling Theory, Bio Statistics, Bioinformatics and Computer Applications (January 05-08, 2011).
3. High Performance Computing: Paradigm and Infrastructure, 2nd Refresher Course on Information and Communication Technology, Academic Staff College , BHU.
4. *Advances in Computational Topology*, Valencia, Spain, June 07-11, 2010 (June 10th).
5. *Computational topology and graph theory*, FAV 2010, Valencia, Spain, June 07-11, 2010 (June 9th).
6. *Applications of topology in communication and sensor networks*, FAV 2010, Valencia, Spain, June 07-11, 2010 (June 8th)
7. *Sorting Algorithms*, CMS Pala, Kochi, April 05-13, 2010.
8. *High Performance Computing* (March 28, 2010), National Conference on Advances and Trends in Web Based Computing(ATWBC-2010), March 28-29, 2010,
9. *Parallel complexity of Graph Algorithms*, (March 29, 2010), National Conference on Advances and Trends in Web Based Computing(ATWBC-2010), March 28-29, 2010,
10. *Distributed Graph Algorithms*(Feb 08), National Seminar on IT enabled Business Practices, SMS, Feb 8-9, 2009, Varanasi.
11. *Parallel sorting algorithms*, (Feb 09), National Seminar on IT enabled Business Practices, SMS, Feb 8-9, 2009, Varanasi.
12. *Parallel Graph Algorithms* (November 17th 2007), National Seminar on Emerging Trends in Information and Communication Technology, 2007.
13. *Asymptotic notations and their importance in algorithms*, (November 18th 2007). National Seminar on Emerging Trends in Information and Communication Technology, 2007.
14. *Roadmap of Information Technology*, 10th International Conference on Information Technology (ICIT), NIT Rourkela, Orissa, Dec 17-20, 2007.
15. *Renormalization and Graph theory*, Renormalization, Max Planck Institute for Mathematics(MPI), Bonn, Dec 15-16, 2006 (Dec 15th).
16. *Quantum computing and Renormalization*, Renormalization, Max Planck Institute for Mathematics(MPI), Bonn, Dec 15-16, 2006 (Dec 16th).
17. *Lee Algorithm and Complexity*, Borel Symposium, ETH Zurich and University of Geneva, June 26- July 06, 2005.
18. *Symbolic algorithms and Lee algebra computation*, Borel Symposium, ETH Zurich and University of Geneva, June 26- July 06, 2005 (June 27th)

19. *Applications of Lie algebra in computing*, University of Geneva, June 26- July 06, 2005 (July 6th)
20. *Some research contribution of Sri Srinivas Ramanujan with Professor G H Hardy*, Workshop on Ramanujan, Birla Institute of Technology, Mesra, December 22, 2002.
21. Significant contribution of Ramanujam in the area of Number Theory, Workshop on Ramunajan, February 22, 2002, Birla Institute of Technology, Mesra.
22. *Computational Singularity theory and its applications*, International Conference in Singularity Theory, University of Sc & Tech, China.
23. Computational aspects of collision singularities, *Computational aspects of classifying singularities*, International Conference in Singularity Theory, University of Sc & Tech, China.
24. *Future system in HPC and Cluster Computing*, FRDCS-2012, MGKVP, Varanasi.
25. *Impact of supercomputing in Sc & engineering*, FRDCS-2012, MGKVP, Varanasi.
26. *High performance data mining algorithms*, FRDCS-2012, MGKVP, Varanasi.

20. Session Chaired:

- International Conference on New Trends in Life Testing, Bayesian Inference, Sampling Theory, Bio Statistics, Bio Informatics and Computer Applications, Banaras Hindu University January 5-8, 2011.
- Research Promotion Workshop on Introduction to Graph and Geometric Algorithms, *Jointly organized by School of Technology and Computer Science, Tata Institute of Fundamental Research Bombay and Banaras Hindu University, January 27-29, 2010.*

20. Professional Recognition:

Membership (Professional Societies) :

- (i)Senior Member, IEEE, IEEE Computer Society, and IEEE Communication Society, (USA).
- (ii)Life Member of High Performance Computing and Networking, Taiwan.
- (iii) Member of [International Society for Engineering Education](#) (IGIP)
- (iv)Life Member Advanced Computing and Communication Society, IISc Bangalore, India.
- (v) Member of the National Academy of Sciences, India. (vi)Life Member of the Indian Mathematical Society, New Delhi.
- (vii) Life Member Indian Society for Technical Education (ISTE), New Delhi.

21. Administrative Expert as a Member/ Chairperson:

- (i) Member of the Academic Council, BHU, Varanasi.
(April 1, 2007- March 31, 2010)
- (ii) Chairman of the Board of Studies, Dept. of Computer Science, BHU, Varanasi (April 1, 2007- March 31-2010)
- (iii) Chairman of the Board of Examiners in Computer Science, BHU Varanasi for two years (April 1, 2007- March 31-2010).
- (iv) Chairman of the Board of Moderators in Computer Science, BHU, Varanasi. (April 1, 2007- March 31-2010).
- (v) Chairman, Ph D Admission Committee, Dept. of Computer Science, BHU –Varanasi (April 1, 2007- March 31-2010).
- (vi) Chairman of the RPC, Dept. of Computer Science, BHU, Varanasi.
(April 1, 2007- March 31-2010)
- (vii) Chairman of the PPC (Policy Planning Committee) Dept. of Computer Science, BHU, Varanasi (April 1, 2007- March 31-2010)
- (viii) Chairman of the DRC, Dept. of Computer Science, BHU, Varanasi (April 1, 2007- March 31-2010)
- (ix) Member of the Board of Studies, Dept. of Computer Science, BHU, Varanasi. (2005, 2006, 2007, 2008, 2009, 2010)
- (x) Member of the Board of Examiners in Computer Science, BHU Varanasi for two years. (2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012)
- (xi) Member of the Board of Moderators in Computer Science, BHU, Varanasi. (2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012)
- (xii) Member of the DRC, Dept. of Computer Science, BHU, Varanasi.
- (xiii) Member of the Board of Moderator, APSU Rewa.
(2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012).
- (xiv) Member of the Faculty Board, Faculty of Science, BHU, Varanasi.
(2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012)
- (xv) Member of the PPC (Policy Planning Committee) Dept. of Computer Science, BHU, Varanasi.
(2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012)
- (xvi) Member of the Steering Committee, Computer Science Courses, South Campus, BHU, Varanasi.
(2005, 2006, 2007, 2008, 2009, 2010)

- (xvii) Member of the DPC, Dept. of Computer Science, BHU, Varanasi.
(2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012)
- (xviii) Member of the Convocation Committee, Faculty of Science, BHU, Varanasi, 2006, 2007, 2008, 2009, 2010, 2011, 2012).
- (xix) Member of the MCA/ MSC syllabus of BHU Varanasi, .
(2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012)
- (xx) Member of the syllabus preparation of PGDCA/CIC, Rajiv Gandhi South Campus, BHU, Varanasi (2005, 2006)
- (xxi) Member committee of Undergraduate Attendance Monitoring, Committee, BHU, Varanasi, 2006, 2007, 2008, 2009, 2010, 2011, 2012.
- (xxii) Expert Member, Ph D Admission Committee, Dept. of Computer Science, BHU -Varanasi.
(2005, 2006, 2007, 2008, 2009, 2010, , 2011, 2012)
- (xxiii) Expert Member of the Meeting of the Selection Committee in Computer Science for Recruitment of Lectures in Computer Science, Mahatama Gandhi Kashi Vidyapeeth, Varanasi.
- (xxiv) Expert Member of the Moderation Committee in Computer Science, MGKVP, Varanasi.
- (xxv) Expert Member of the Board of Studies in Computer Science, MGKVP, Varanasi.
- (xxvi) Expert Member of the Board of studies in Computer Science, Chitrakoot Gramin Vishwavidyalaya, Chitrakoot, Satana(MP).
- (xxvii) Expert Member of the Board of studies in Computer Science, IGNOU, New Delhi.
- (xxviii) Expert Member of the Board of Examiners in Computer Science, UPSC, Haridwar (Uttarakhand).
- (xxix) Member of the RDCU, Banaras Hindu University, Varanasi.
- (xxx) Expert Member of the Board of Examiners in Computer Science, VYAPM, Bhopal (MP).
- (xxxi) External Expert Member of the Doctoral Committee in Computer Science & Engineering, Birla Institute of Technology, Mesra, Patna Campus since 2006.

(xxxii) Expert Member of the RPC of 6 research students of Department of Mathematics, BHU since 2007.

(xxxiii) Member of the faculty Exhibition Committee, Alumni Meet- Dec 2012, BHU.

(xxxiv) Member of the Convocation Organizing Committee-2011, BHU.

(xxxv) Member of the Convocation Organizing Committee-2012, BHU

22. Important Extra Curricular Activities:

(i). Secretary, Student Union, Govt. College, Mouganj, (India) 1987-1988. (ii). NSS, 1988-1990.

(iii). Frequent Broadcast of popular talks on Computer Science and Technology related topics by all India Radio.

(iv). Publication of popular articles in local daily News paper.

(P. K. Mishra)